Faheem Nizar

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Research Interests

3D scene understanding, embodied computer vision and robot learning, with a focus on learning geometric and semantic representations of dynamic environments from images and video. Interested in leveraging these with Agentic AI planning and safety first robot interactions.

Education

Indian Institute of Technology (IIT) Kanpur Kanpur, India
Bachelor of Technology in Electrical Engineering; Minor in Machine Learning *Nov. 2020 – May 2024*

- **GPA**: **8.6/10** | Distinction for Academic Performance (2024); All India Rank 1136 in JEE Advanced 2020
- **Coursework**: Machine Learning, Big Data Visual Analytics, Image Processing, Computer Graphics, Data Structures & Algorithms, Probability & Statistics

Publications

Peer-reviewed journals

E. Lumer, A. Gulati, **F. Nizar**, D. Hedroits, A. Mehta, H. Hwangbo, V. K. Subbiah, P. Basavaraju and J. Burke, "Tool and Agent Selection for Large Language Model Agents in Production: A Survey," *IEEE Conference on Artificial Intelligence*, May 2026.

F. Nizar, E. Lumer, A. Gulati, P. Basavaraju and V. K. Subbiah, "Agent-as-a-Graph: Knowledge Graph-Based Tool and Agent Retrieval for LLM Multi-Agent Systems," *ICAART 26*, Mar. 2026. *(first author)*

S. Tripathi, **F. Nizar**, O. J. Pandey, T. Sandhan and R. M. Hegde, "ENADL: Towards Performance Improvement of IoT Networks Using Deep Learning-Based Node Fault Prediction," *IEEE Transactions on Reliability*, Mar. 2025.

S. Dutta, **F. Nizar**, A. Amaan and A. Acharya, "Visual Analysis of Prediction Uncertainty in Neural Networks for Deep Image Synthesis," *IEEE Transactions on Visualization and Computer Graphics*, May 2024.

Preprints

E. Lumer, **F. Nizar**, A. Jangiti, K. Frank, A. Gulati, M. Phadate and V. K. Subbiah, "Don't Break the Cache: An Evaluation of Prompt Caching for Long-Horizon Agentic Tasks," arXiv preprint, Jan. 2026.

E. Lumer, **F. Nizar**, A. Gulati, P. Basavaraju and V. K. Subbiah, "Tool-to-Agent Retrieval: Bridging Tools and Agents for Scalable LLM Multi-Agent Systems," arXiv preprint, Nov. 2025.

Research Experience

Indian Institute of Technology Kanpur Kanpur, India
Undergraduate Researcher | Advisor: Prof. Tushar Sandhan *Jan 2024 – Jun 2024*

- Built high-fidelity simulations of large-scale IoT network communication in **NetworkX**, injecting physical node and link faults to reproduce real-world system failures
- Proposed an **Edge-and-Node Accumulation** graph deep learning architecture that learns spatio-topological signatures of faulty network behavior, reaching **98%** node-fault prediction accuracy
- Designed centrality-based resource allocation that extended network lifetime by **43%** through load redistribution
- Work published as **ENADL** in *IEEE Transactions on Reliability* (2025)

Indian Institute of Technology Kanpur Kanpur, India
Undergraduate Researcher | Advisor: Prof. Soumya Dutta *Jan 2023 – Mar 2023*

- Studied prediction uncertainty and confidence estimation in deep networks used for **deep image synthesis** on scientific datasets
- Compared **Ensemble** learning against **Monte Carlo Dropout** to quantify epistemic uncertainty, and identified the sample and ensemble counts needed for stable image generation
- Built visualization pipelines (parallel coordinate plots, 2D heatmaps) for multidimensional analysis of uncertainty metrics
- Found ensemble methods more robust than Monte Carlo Dropout for uncertainty quantification; published in *IEEE Transactions on Visualization and Computer Graphics* (2024)

Indian Institute of Technology Kanpur Kanpur, India
Research Intern, Antenna Tracker | Supervisor: Prof. Abhishek K. Gupta *May 2022 – Aug 2022*

- Built a real-time antenna tracking system to hold long-range, low-latency links between UAVs and a ground station
- Set up **MAVLink** bidirectional telemetry over ESP-NOW on ESP32, and a self-healing **IEEE 802.11s** BATMAN mesh on Raspberry Pi nodes for swarm UAV communication
- Engineered **PWM-to-PPM** conversion to drive stepper motors from a Pixhawk flight controller for precise actuation
- Integrated a **YOLOv5** visual tracking module to detect and follow UAVs in real time for closed-loop antenna alignment for short ranges

Selected Projects & Competitions

GenCast Hurricane Forecasting | *FutureSprint Hackathon* Oct 2025

- Ran the **GenCast** mini ensemble model to predict **15-day** hurricane track ensembles from ECMWF weather data
- Mapped predicted tracks to affected zip codes for targeted warnings, and generated stakeholder reports using a grounded web-search agent

IoT Innovation Contest | *FutureThink Cloud Innovation Summit* Sep 2024

- Won **first place** across all Indian PwC offices; led a team of 5 through prototype development
- Integrated generative AI into IoT kiosks using Raspberry Pi, OpenCV face detection, and a conversational assistant

Technical Skills

Languages: Python, C++, C, SQL, JavaScript
ML / DL: PyTorch, PySpark, Hugging Face Transformers, XGBoost, MLflow
Vision & graphs: OpenCV, YOLO, NetworkX
LLM / agents: LangChain, LangGraph
Systems & tools: Linux, Docker, Databricks, Git, Flask, Streamlit

Professional Experience

PwC US Bangalore, India
Senior Associate, Data Scientist *Jul 2024 – Present*

- Built a **spatio-temporal** motion generation system that synthesizes 3D character motion from text prompts for a gaming client, cutting motion-capture cost for NPC animation
- Developed an AI Co-Analyst microservice (LangGraph) for tax anomaly detection, backed by a **NetworkX** causal knowledge graph of **200+** SME-approved relationships to surface root-cause hypotheses
- Trained and productionized a **DistilBERT** classifier on **Databricks** with MLflow lifecycle management, cutting a benchmarking task from **4 weeks to 10 minutes**
- Built a semantic **LLM routing** layer that scores query difficulty and directs traffic to the cheapest capable model, reducing API inference cost by **40%**

American Express Bangalore, India
Data Analyst Intern (Pre-Placement Offer) *May 2023 – Jul 2023*

- Built a credit-default risk framework for low-tenure, high-balance cardholders by engineering risk variables and training an **XGBoost** model on internal and external tradeline data
- Ran large-scale segment simulations for a **2%** GINI gain and **4%** higher default capture, with 3 of 4 engineered features in the top 100 by model importance

Leadership & Service

PwC US Bengaluru, India
Learning & Development Lead, Data Science Team *Jul 2024 – Present*

- Ran internal sessions on prompt engineering and agentic AI, and managed staffing across a **46-member** team

Academics and Career Council, IIT Kanpur Kanpur, India
Head of Web Development *May 2022 – May 2023*

- Led a team of **6** developers building portals used across the campus community